MODULE 4 OF 6

Spatial Continuity and Variography

The heart of geostatistics: measuring, modelling and exploiting spatial dependence in grade

Classical statistics treats samples as independent — but grade samples are not independent: two pits 50 m apart tell you more about each other than two pits 5 km apart. Variography is the mathematics that measures, models and exploits that spatial dependence. Every estimate in Module 5 is built on the variogram fitted here.

LEARNING OBJECTIVES

- Explain the regionalised variable and the random function model.
- State the second-order and intrinsic stationarity hypotheses.
- Define the covariance and the semivariogram and derive the relationship between them.
- Compute an experimental variogram and read its nugget, sill and range.
- Fit a licit variogram model and recognise anisotropy.
- Build and interpret the Toro SnO_2 variogram.

1 The regionalised variable and the random function

Grade is a number attached to a location, written $z(\mathbf{x})$. Matheron called such a quantity a **regionalised variable** — distributed in space with a *structured* character (reflecting geology) and a *random* character (point-to-point fluctuation no deterministic equation predicts).

We have exactly one grade value at each sampled point — one realisation of reality. To speak of probability and variance we model that single reality as a **random function**: at every location $\mathbf{x}$, treat the grade $Z(\mathbf{x})$ as a random variable; the whole collection is a random function. Our measured map is one realisation drawn from an imagined ensemble of geologically possible deposits.

2 Stationarity hypotheses

We have one realisation but need ensemble averages. Stationarity is the bridge: it lets us replace ensemble averages with spatial averages over our single dataset.

Second-order (weak) stationarity requires a constant mean and a covariance depending only on the separation vector $\mathbf{h}$:

$$\mathbb{E}[Z(\mathbf{x})] = m, \qquad \mathrm{Cov}\big[Z(\mathbf{x}), Z(\mathbf{x}+\mathbf{h})\big] = C(\mathbf{h}) \tag{4.1–4.2}$$

Setting $\mathbf{h} = 0$ gives $C(0) = \mathrm{Var}[Z(\mathbf{x})]$, the variance of the random function. Second-order stationarity requires a finite $C(0)$. Some geological variables have variance that grows without limit as the domain expands, and for them geostatistics relies on a weaker assumption.

The intrinsic hypothesis

The intrinsic hypothesis requires stationarity only of the *increments* $Z(\mathbf{x}+\mathbf{h}) - Z(\mathbf{x})$:

$$\mathbb{E}[Z(\mathbf{x}+\mathbf{h}) - Z(\mathbf{x})] = 0, \quad \mathrm{Var}[Z(\mathbf{x}+\mathbf{h}) - Z(\mathbf{x})] = 2\gamma(\mathbf{h})$$

Every second-order stationary function is intrinsic but not the reverse. This is the working assumption of practical geostatistics.

3 The semivariogram and its link to covariance

The **semivariogram** $\gamma(\mathbf{h})$ is the central tool of geostatistics — half the variance of the increment:

$$\gamma(\mathbf{h}) = \tfrac{1}{2}\,\mathrm{Var}[Z(\mathbf{x}+\mathbf{h}) - Z(\mathbf{x})] = \tfrac{1}{2}\,\mathbb{E}\Big[\big(Z(\mathbf{x}+\mathbf{h}) - Z(\mathbf{x})\big)^2\Big] \tag{4.3–4.4}$$

Small γ means values stay alike across $\mathbf{h}$; large γ means they decorrelate.

Expanding the square in (4.4):

$$2\gamma(\mathbf{h}) = \mathbb{E}[Z(\mathbf{x}+\mathbf{h})^2] + \mathbb{E}[Z(\mathbf{x})^2] - 2\mathbb{E}[Z(\mathbf{x}+\mathbf{h})Z(\mathbf{x})]$$

For any random variable, $\mathbb{E}[Z^2] = C(0) + m^2$. And $\mathbb{E}[Z(\mathbf{x}+\mathbf{h})Z(\mathbf{x})] = C(\mathbf{h}) + m^2$. Substituting:

$$2\gamma(\mathbf{h}) = 2C(0) + 2m^2 - 2C(\mathbf{h}) - 2m^2 = 2\big(C(0) - C(\mathbf{h})\big)$$

$$\boxed{\gamma(\mathbf{h}) \;=\; C(0) - C(\mathbf{h})}$$

At $\mathbf{h} = 0$, $\gamma = 0$: a value is identical to itself. As $\mathbf{h}$ grows, $C(\mathbf{h})$ decays towards zero, so γ *rises* towards $C(0)$, the variance. The variogram is the covariance turned upside down.

4 The experimental variogram

From data we estimate (4.4) with the Matheron estimator:

$$\hat{\gamma}(\mathbf{h}) = \frac{1}{2\,N(\mathbf{h})} \sum_{i=1}^{N(\mathbf{h})} \big[z(\mathbf{x}_i) - z(\mathbf{x}_i + \mathbf{h})\big]^2 \tag{4.5}$$

Real samples are never separated by exactly $\mathbf{h}$, so pairs are collected into **lag bins** defined by a lag and a tolerance. Three rules: each lag should carry at least about 30 pairs (below 20 is noise); interpret only out to about half the field extent; the lag spacing should be near the typical data spacing.

5 Anatomy of the variogram

A typical variogram rises and flattens. Three parameters describe it.

Nugget C_0 — the apparent jump at vanishingly small h. $\gamma(0) = 0$ in theory; a positive intercept means two samples nearly at the same place still differ. It is micro-scale geological variability plus Module 2's sampling-and-assay error. Pure unstructured noise.

Sill $C_0 + C$ — the plateau. Under the boxed result of section 4.3 the sill equals the variance $C(0)$ of the data. C alone is the structured part of the variance.

Range a — the lag at which the variogram reaches the sill; the distance of influence. Beyond the range, two samples are statistically independent.

The most diagnostic single number is the **relative nugget**:

$$\text{relative nugget} = \frac{C_0}{C_0 + C} \tag{4.6}$$

Below 0.25: well-structured. 0.25–0.5: moderate noise. Above 0.5: highly erratic — half the variability is unstructured.

6 Variogram models

The experimental variogram is known only at noisy, discrete lags. Kriging needs $\gamma(\mathbf{h})$ everywhere — and the function must be **conditionally negative semi-definite** so the kriging system has a unique solution and never returns negative variance. So we fit one of a small library of *licit* models.

Spherical — the workhorse

$$\gamma(h) = \begin{cases} C_0 + C\left[\dfrac{3h}{2a} - \dfrac{1}{2}\left(\dfrac{h}{a}\right)^3\right], & 0 < h \le a \\ C_0 + C, & h > a \end{cases} \tag{4.7}$$

Reaches the sill exactly at the range a, linear near the origin. Used for the Toro estimate.

Exponential

$$\gamma(h) = C_0 + C\left[1 - \exp\left(-\tfrac{3h}{a}\right)\right] \tag{4.8}$$

Approaches the sill asymptotically; a is the practical range where 95% of the sill is reached. Suits erratic mineralisation.

Gaussian

$$\gamma(h) = C_0 + C\left[1 - \exp\left(-\tfrac{3h^2}{a^2}\right)\right] \tag{4.9}$$

Parabolic at the origin: describes extremely continuous properties (thicknesses, weathering surfaces). Use with care — numerically unstable without a nugget.

Power — unbounded

$$\gamma(h) = C_0 + p\,h^{\omega}, \qquad 0 < \omega < 2 \tag{4.10}$$

No sill, no range. Describes a variable that decorrelates without limit.

A sum of licit models is itself licit. Real deposits often need a **nested** variogram — a nugget plus a short-range structure (local mineralisation) plus a long-range structure (regional trend).

7 Anisotropy

Grade continuity is rarely the same in every direction. A placer is more continuous *along* its palaeochannel than across it. Variograms are computed **directionally** — pairs binned not only by distance but by the azimuth of the vector joining them, within an angular tolerance.

When directional variograms share a sill but differ in range, the structure has **geometric anisotropy**, described by an ellipse (major axis along the direction of greatest continuity). When the sill itself changes with direction, it is **zonal anisotropy**. A **variogram map** computes $\hat{\gamma}$ over a 2D grid of lag vectors; its elliptical contours reveal both strength and orientation of anisotropy at a glance.

8 The Toro variogram, built and fitted

The variogram is computed on the 26 composited Toro pit grades, with a 180 m lag matching the typical Bishiwai pit spacing. Fitting follows three defensible decisions: only the six lags carrying at least 20 pairs drive the fit (the three sparse outer lags are noise); the sill is fixed to the sample variance (the boxed result of section 4.3); and an isotropic spherical model is adopted because the directional data cannot resolve anisotropy.

Parameter	**Value**	**Interpretation**
Nugget C_0	1.88	unstructured noise — sampling + micro-variability
Sill $C_0 + C$	3.44	total variance of composited grades
Range a	1052 m	distance of grade influence
Relative nugget	0.55	more than half the variance is unstructured

A relative nugget of 0.55 means over half the grade variance has no spatial structure the data can resolve. Some is the genuine erratic character of coarse cassiterite; much is the fundamental sampling error of panned-concentrate sampling. The structured range of about 1 km is encouraging — drainage-scale continuity — but a high-nugget variogram is a quantitative warning: this data will estimate weakly. Lowering the nugget is a sampling problem (Module 2), not an estimation problem.

```
import numpy as np
```

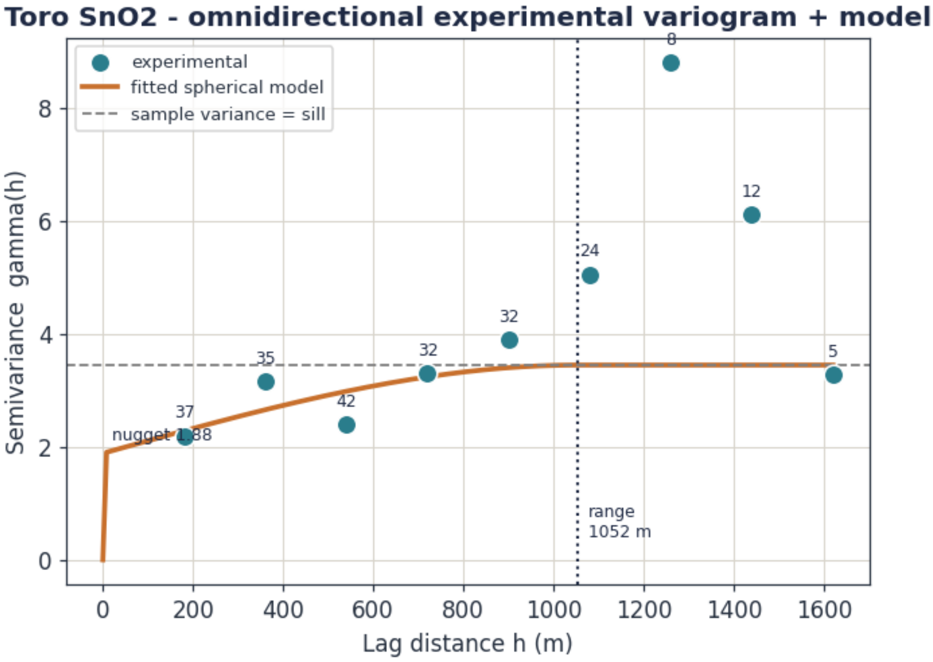

Figure 1 *– Toro SnO_2 omnidirectional experimental variogram with fitted spherical model. Pair counts beside each point.*

```
def experimental_variogram(x, y, z, lag, nlags, tol=None):
    """Matheron estimator - equation (4.5)."""
    if tol is None: tol = lag / 2.0
    n = len(z)
    dx = x[:, None] - x[None, :]
    dy = y[:, None] - y[None, :]
    h  = np.sqrt(dx**2 + dy**2)
    diff2 = (z[:, None] - z[None, :])**2
    iu = np.triu_indices(n, k=1)
    h, diff2 = h[iu], diff2[iu]
    out = []
    for k in range(1, nlags + 1):
        centre = k * lag
        m = np.abs(h - centre) <= tol
        if m.sum() >= 2:
            out.append((centre, 0.5 * diff2[m].mean(), int(m.sum())))
    return out

def spherical(h, nugget, sill, rng):
    """Licit spherical model - equation (4.7)."""
    h = np.asarray(h, float)
    g = np.where(h <= rng,
                 nugget + (sill - nugget) * (1.5*h/rng - 0.5*(h/rng)**3),
                 sill)
    return np.where(h == 0, 0.0, g)
```

Module 4 of the Toro Resource Estimation course. Next: Module 5, Grade Estimation.